

Data Analysis Report — Operations Challenge, canning
lines 14 · 17 · 19 at El Prat factory.

Audience: hackathon team · **Source:** 2,142 production OFs in 2025 across 3 lines · Database: `db/linewise.duckdb`

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1. Executive summary

2,142 Production OFs analysed (excl. LIMPIEZA)	2,792,478 Hectolitres produced (HL) 2025	49.1% Weighted-average OEE across L14-17-19	10.7 pts OEE drop when a size change occurs
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One-line headline

THE OPPORTUNITY
L14 averages **42%** OEE, L17 **53%**, L19 **48%**. The single biggest OEE killer is a **volume size change** (1/3 ↔ 1/2): **−10.7** OEE points on average. Even +1 OEE point across the three lines = millions of additional cans per year.

Top three actionable findings

1. **Cluster same-size runs.** Size changes (1/3 ↔ 1/2) are the headline OEE killer. Sequencing should avoid them whenever feasible. (See §4.)
2. **The Tiempo file gives us free explainability.** Lost time already decomposed into CIP, baja velocidad, saturación, falta de producto, etc. The diagnostics panel can use these categories natively without inventing them. (See §6.)
3. **Sub-portfolio of chronic underperformers** (DAMM LEMON 4-pack, FREE DAMM 4-pack, COMLOT, SKOL 12-pack): mean OEE 23–35%. Worth flagging as a category in the UI. (See §8.)

2. Damm data model — formulas & relationships

Encoded verbatim from the data-model diagram provided by Damm. These are also stored as queryable tables (`_meta_formulas` , `_meta_relationships`) inside the DuckDB database.

Formulas

concept	formula	notes
OEE	OEE = Disponibilidad * Rendimiento * C...	Multiplicative. Available in raw_oeef...
Disponibilidad	Disponibilidad = Tiempo_de_funcionamie...	Already provided per OF in raw_oeedis...
Rendimiento	Rendimiento = (Tiempo_ciclo_ideal * Pr...	Already provided per OF in raw_oeeren...
Calidad	Calidad = Produccion_buena / Produccio...	Implicit in OEE; equals 1.0 when no re...
Tiempo cambio (real)	Tiempo_cambio = PAR_TOT - (PNP + LIMPI...	fact_runs.horas_cambio. Use this (not ...
IDLE	IDLE no afecta OEE	Exclude IDLE from OEE-impact arithmeti...

CRITICAL CORRECTION

Earlier we used `horas_cip` as the proxy for changeover duration per OF. The correct Damm formula is:

$$\text{Tiempo cambio} = \text{PAR_TOT} - (\text{PNP} + \text{LIMPIEZA} + \text{IDLE})$$

This is now exposed as `fact_runs.horas_cambio`. **IDLE does NOT affect OEE** — keep it informational, exclude from impact arithmetic.

Entity relationships

The central nexus is **MES / OF (Material × Production Order)**. Every analytical table joins to it.

entity_from	entity_to	semantics	edge_style
MES_OF	CAMBIOS	central nexus -> changeover events per...	blue dashed
MES_OF	OEE	central nexus -> OEE master	blue dashed
MES_OF	TIEMPO	central nexus -> lost-time breakdown	blue dashed
MES_OF	VOLUMEN	central nexus -> produced volumes	blue dashed
MES_OF	TIEMPO_CAMBIO_TEORICO	central nexus <-> theoretical changeov...	red bidirectional (CRITICAL)
CAMBIO_FORMATO	TIEMPO_CAMBIO_TEORICO	format change matrix feeds theoretical...	red dashed (CRITICAL)
MANTENIMIENTO_LIMPIEZAS	TIEMPO_CAMBIO_TEORICO	cleaning/maintenance events feed theor...	red dashed (CRITICAL)
CAMBIOS	TIPO_DE_CAMBIO	each changeover event has a type (Cont...	green
TIPO_DE_CAMBIO	PREVIOUS_OF	WARNING: tipo de cambio requires inspe...	green (CRITICAL warning)
OEE	CAMBIO_SI_NO	OEE row tags whether a changeover occu...	green
OEE	PCT_OEE_PCT_RENDIMIENTO	OEE row reports % OEE and % Rendimiento	green
TIEMPO	H_TOT	tiempo container -> total hours	green
TIEMPO	PAR_TOT_1	tiempo container -> total stoppage (co...	green
TIEMPO	PNP_2	tiempo container -> planned non-produc...	green
TIEMPO	LIMPIEZA_3	tiempo container -> cleaning time (com...	green
TIEMPO	IDLE_4	tiempo container -> idle time (compone...	green
TIEMPO	PCT_DISP_CALID_REND	tiempo container -> the three OEE comp...	green
VOLUMEN	HL	volumen container -> hectolitres	green
VOLUMEN	UDS	volumen container -> units	green
VOLUMEN	OEE	volumen container -> redundant OEE for...	green

3. Data inventory

Raw tables (lossless ingestion)

table_name	source_file	rows	n_cols	description
raw_actual_2026	Produccion_L14,17,19_18-22.xlsx	36	20	Actual production for May 18-22, 2026 ...
raw_cambios	Cambios 14_17_19_ 2025.xlsx	2,181	31	Changeover events per OF. C. PRINCIPAL...
raw_cf_lata_barril	Tabla CF Prat 2026_14_17_19.xlsx	46	8	Theoretical changeover matrix LATA/BAR...
raw_cf_tiempos_adicionales	Tabla CF Prat 2026_14_17_19.xlsx	37	12	Additional cleaning/maintenance time r...
raw_data_extra	data - 2026-05-18T181640.542.xlsx	2,276	45	Additional OEE history (Sep-Oct 2025 o...
raw_diario_hl_planif	Diario HI_Planif.xlsx	44	100	Daily HL planning report May 18-24 202...
raw_diario_hl_planif_headers	Diario HI_Planif.xlsx	98	3	Sidecar with original multi-line Excel...
raw_mantenimiento	Mantenimiento 14_17_19_ 2025.xlsx	2,276	25	Maintenance events per OF + standalone...
raw_oeo	OEE 14_17_19_ 2025.xlsx	2,274	45	OEE master per OF, 43 cols of product ...
raw_plan_2026	Planificado - producciones 14 - 17 - 1...	78	19	Blue Yonder / JDA theoretical plan for...
raw_plan_2026_v2	Planificado - producciones 14 - 17 - 1...	78	18	Refreshed Planificado for May 2026 (sa...
raw_tiempo	Tiempo 14_17_19_ 2025.xlsx	2,278	35	Per-OF lost-time breakdown: CIP, baja ...
raw_volumen	Volumen 14_17_19_ 2025.xlsx	2,278	20	Per-OF produced volumes (UDS, HL).

Canonical / derived tables

table_name	rows	description
_meta_formulas	6	Damm OEE & changeover formulas extract...
_meta_relationships	20	Edges from Damm's data-model diagram (...)
dim_line	3	Production lines (L14/L17/L19) and the...
dim_sku	170	SKU master with product hierarchy and ...
dim_theoretical_changeover_matrix	86	Theoretical changeover minutes per (li...
fact_changeovers	2,180	Per-OF transitions on the same line wi...
fact_diario_hl_planif	197	Daily HL planning long-format (linea, ...
fact_limpieza	133	Standalone LIMPIEZA / cleaning WOs.
fact_lost_time	19,602	Long-format lost-time per OF and categ...
fact_plan_vs_actual_2026	91	May 2026 Blue Yonder plan aligned with...
fact_runs	2,184	Canonical per-OF run: OEE + lost-time ...

4. OEE baseline per line

Aggregated over all production OFs in 2025, excluding LIMPIEZA WOs and outlier rows where **H. Tot. > 100h**.

linea	n_ofs	oee_media	oee_mediana	oee_std	oee_p10	oee_p25	oee_p75	oee_p90	disp_media	rend_media	inef_media	hl_total	uds_total
14.00	428.00	0.4216	0.4471	0.1311	0.2388	0.3255	0.5224	0.5654	0.5592	0.7561	0.1376	566,900.5	178,291,181.0
17.00	934.00	0.5317	0.5416	0.1503	0.3356	0.4384	0.6379	0.7116	0.5997	0.8799	0.0680	1,128,150.7	348,838,277.0
19.00	780.00	0.4803	0.4748	0.1674	0.2658	0.3556	0.5959	0.7132	0.5915	0.7993	0.1112	1,097,426.4	301,125,470.0

READ

L17 is the strongest line by mean OEE, L14 the weakest. The p10–p90 spread is wide on every line (L19: 24 % → 71 %), confirming that *which* SKU runs and *after what* matter a lot. That spread is the optimization opportunity LineWise exploits.

5. What changeovers cost — by type & dimension

By C. PRINCIPAL (the type label)

cambio_tipo_principal	n	oe_medio	oe_mediano	disp_media	rend_media
Contenido	756	0.5046	0.5170	0.5880	0.8528
Pack. Secundario	701	0.4777	0.4833	0.5755	0.8201
Palet	207	0.5504	0.5434	0.6612	0.8269
Pack, Primario	127	0.4209	0.4180	0.5329	0.7810
Volumen Envase	109	0.3897	0.3592	0.5098	0.7534
Referencia	96	0.5400	0.5425	0.6525	0.8141
Marca	72	0.4739	0.4677	0.5889	0.8045
-2	64	0.5249	0.5363	0.6356	0.8191
Tapa/Tapón	12	0.5486	0.5147	0.6790	0.8057

By line x type

linea	cambio_tipo_principal	n	oe_medio	oe_mediano
14	Pack. Secundario	186	0.3894	0.4060
14	Contenido	78	0.4566	0.4719
14	Palet	72	0.4472	0.4560
14	Referencia	30	0.4750	0.4788
14	-2	29	0.4770	0.5153
14	Pack, Primario	26	0.3624	0.3515
14	Marca	4	0.4602	0.4848
14	Tapa/Tapón	4	0.4516	0.5020
14	Volumen Envase	2	0.2479	0.2479
17	Contenido	429	0.5246	0.5360
17	Pack. Secundario	345	0.5267	0.5343
17	Palet	67	0.5852	0.5669
17	Marca	38	0.5060	0.5045
17	Referencia	30	0.5928	0.6197
17	-2	23	0.5404	0.5437
17	Tapa/Tapón	1	0.3328	0.3328
19	Contenido	249	0.4852	0.4950
19	Pack. Secundario	170	0.4748	0.4712
19	Volumen Envase	107	0.3923	0.3592
19	Pack, Primario	101	0.4360	0.4308
19	Palet	68	0.6255	0.6571
19	Referencia	36	0.5501	0.5499
19	Marca	30	0.4352	0.4374
19	-2	12	0.6110	0.6191
19	Tapa/Tapón	7	0.6348	0.6472

OEE delta when each change-dimension flips

dimension	oeesin cambio	oeesin cambio	delta	n sin cambio	n sin cambio
volum	0.3897	0.4963	-0.1066	109.00	2,035.0
producto	0.4692	0.5276	-0.0584	1,349.0	795.00
brand	0.4704	0.5275	-0.0570	1,377.0	767.00
cap	0.4395	0.4949	-0.0555	158.00	1,986.0
primario	0.4661	0.5055	-0.0393	796.00	1,348.0
secundario	0.4692	0.5030	-0.0339	772.00	1,372.0
palet	0.4754	0.5011	-0.0258	856.00	1,288.0
envase	nan	0.4908	nan	0.00	2,144.0

WHY THIS MATTERS

Size (volum) changes are the single biggest OEE drag (–10.7 pts). They’re relatively rare (~109 events in 2025), but each one is expensive. **Producto, brand** and **CAP (tapón)** changes are next: –5 to –6 pts. *Palet* changes hurt least (–2.6 pts). The sequencer should weight these accordingly when choosing among candidates.

6. Real vs theoretical changeover times (Damm formula)

Using the corrected formula $\text{horas_cambio} = \text{PAR_TOT} - (\text{PNP} + \text{LIMPIEZA} + \text{IDLE})$ per OF, compared against the theoretical reference in `dim_theoretical_changeover_matrix`. We also retain the earlier CIP-only view for transparency.

A18 — Damm-formula version (per OF, in-OF time only)

linea	from_estado	to_estado	n	cambio_real_min_medio	cambio_real_min_mediano	cip_min_medio	teorico_min	delta_real_vs_teorico	oee_medio
19	2/5	1/2	5	27.30	32.70	4.00	360	-332.80	0.4265
19	1/3	1/2	47	21.20	14.40	0.1000	360	-338.80	0.3967
19	1/3	2/5	33	21.10	16.20	1.5000	360	-338.90	0.3938
19	1/2	1/3	44	18.30	2.2000	8.5000	360	-341.70	0.4635
19	2/5	1/3	35	15.90	11.70	3.8000	360	-344.10	0.4924
19	1/2	2/5	8	5.8000	0.00	0.00	360	-354.20	0.4329

A04 — CIP-only proxy (legacy view)

linea	from_estado	to_estado	n	cip_real_min_medio	teorico_min	delta_min	oee_medio	cip_total_min
19	1/2	1/3	44	8.5000	360	-351.50	0.4635	372.40
19	2/5	1/2	5	4.00	360	-356.00	0.4265	19.90
19	2/5	1/3	35	3.8000	360	-356.20	0.4924	134.00
19	1/3	2/5	33	1.5000	360	-358.50	0.3938	49.70
19	1/3	1/2	46	0.1000	360	-359.90	0.3974	3.9000
19	1/2	2/5	7	0.00	360	-360.00	0.4149	0.00

IMPORTANT NUANCE

The Damm-formula *horas_cambio* captures only the in-OF portion of a changeover. The full changeover budget = in-OF *horas_cambio* + duration of any standalone LIMPIEZA WO(s) that occurred between the previous and current production OF (see §11). The L19 1/3 ↔ 1/2 size change averages 6 h theoretical but only ~21 min of in-OF changeover time — the bulk happens as separate LIMPIEZA WOs.

7. Anatomy of PAR_TOT per line

Where does the total stoppage time per OF actually go? Average minutes per OF, decomposed.

linea	n_ofs	par_tot_min_avg	pnp_min_avg	limpieza_in_of_min_avg	idle_min_avg	cambio_real_min_avg	cip_min_avg
14.00	436.00	346.30	318.80	7.5000	10.10	9.9000	4.2000
17.00	950.00	159.80	131.70	8.7000	5.3000	14.10	2.2000
19.00	792.00	172.60	129.50	9.5000	17.00	16.70	1.9000

L14 IS DOMINATED BY PNP

L14 has an average **319 min** of PNP (Planned Non-Production) per OF — vs ~131 min on L17 and L19. PNP is shorter shifts / scheduled downtime, not lost productivity. This explains a big chunk of L14's low OEE: less wall-clock running time per OF.

8. Lost-time decomposition

From `fact_lost_time` . *marcha* (productive) excluded.

linea	categoria	n_ofs	horas_totales	minutos_medios_por_of	minutos_medianos_por_of
14	paro_maquina	436	2,503.5	344.50	227.30
14	pnp	436	2,316.9	318.80	195.30
14	saturacion_salida	436	1,342.0	184.70	109.20
14	baja_velocidad	436	1,032.2	142.10	66.20
14	idle	436	73.30	10.10	0.00
14	cip	436	30.60	4.2000	0.00
14	falta_producto	436	22.50	3.1000	0.5000
14	esterilizacion	436	9.5000	1.3000	0.00
17	paro_maquina	950	2,561.3	161.80	121.30
17	pnp	950	2,084.9	131.70	88.20
17	saturacion_salida	950	744.90	47.00	18.90
17	baja_velocidad	950	575.00	36.30	18.40
17	falta_producto	950	219.20	13.80	6.9000
17	idle	950	84.50	5.3000	0.00
17	esterilizacion	950	56.70	3.6000	0.00
17	cip	950	35.10	2.2000	0.00
19	paro_maquina	792	2,151.2	163.00	115.70
19	pnp	792	1,709.0	129.50	85.80
19	baja_velocidad	792	1,002.4	75.90	33.30
19	saturacion_salida	792	573.10	43.40	21.00
19	idle	792	224.70	17.00	0.00
19	falta_producto	792	204.70	15.50	8.1000
19	esterilizacion	792	40.40	3.1000	0.00
19	cip	792	25.70	1.9000	0.00

USE THIS DIRECTLY IN THE UI

The brief asks us to *explain* why OEE drops. We don't need to invent categories — the Damm Tiempo file already provides them. Power the diagnostic panel and Claude's tool responses with these eight categories natively.

9. Temporal patterns

OEE by day of week (1=Mon, 7=Sun)

linea	dia_semana	n	oeo_medio
14.00	1.00	63.00	0.4121
14.00	2.00	92.00	0.4187
14.00	3.00	65.00	0.4112
14.00	4.00	64.00	0.4136
14.00	5.00	70.00	0.4168
14.00	6.00	54.00	0.4459
14.00	7.00	20.00	0.4746
17.00	1.00	141.00	0.5048
17.00	2.00	176.00	0.5336
17.00	3.00	135.00	0.5439
17.00	4.00	133.00	0.5233
17.00	5.00	138.00	0.5676
17.00	6.00	130.00	0.5371
17.00	7.00	81.00	0.4979
19.00	1.00	101.00	0.4784
19.00	2.00	149.00	0.4934
19.00	3.00	147.00	0.4375
19.00	4.00	163.00	0.4420
19.00	5.00	106.00	0.5059
19.00	6.00	73.00	0.5563
19.00	7.00	41.00	0.5424

OEE by month

linea	anyo	mes	n	oeo_medio	hl
14.00	2,025.0	1.00	37.00	0.3804	32,283.3
14.00	2,025.0	2.00	40.00	0.4449	45,718.8
14.00	2,025.0	3.00	19.00	0.3907	29,424.6
14.00	2,025.0	4.00	28.00	0.4423	48,610.3
14.00	2,025.0	5.00	43.00	0.4219	48,575.0
14.00	2,025.0	6.00	39.00	0.4505	51,328.2
14.00	2,025.0	7.00	46.00	0.3634	52,669.5
14.00	2,025.0	8.00	31.00	0.4262	47,684.0
14.00	2,025.0	9.00	31.00	0.4296	60,691.7
14.00	2,025.0	10.00	47.00	0.4814	68,773.7
14.00	2,025.0	11.00	35.00	0.4126	47,303.7
14.00	2,025.0	12.00	32.00	0.3981	33,837.7
17.00	2,025.0	1.00	62.00	0.5240	71,410.7
17.00	2,025.0	2.00	73.00	0.5761	98,466.9
17.00	2,025.0	3.00	94.00	0.5572	115,365.6
17.00	2,025.0	4.00	77.00	0.5093	97,248.8
17.00	2,025.0	5.00	102.00	0.5275	103,032.9
17.00	2,025.0	6.00	92.00	0.5153	106,346.0
17.00	2,025.0	7.00	83.00	0.4600	99,588.7
17.00	2,025.0	8.00	85.00	0.5280	111,822.9
17.00	2,025.0	9.00	71.00	0.5582	102,861.2
17.00	2,025.0	10.00	64.00	0.5189	73,434.1
17.00	2,025.0	11.00	72.00	0.5528	78,490.3
17.00	2,025.0	12.00	59.00	0.5688	70,082.5
19.00	2,025.0	1.00	72.00	0.5021	68,118.8
19.00	2,025.0	2.00	63.00	0.4999	90,547.2
19.00	2,025.0	3.00	57.00	0.4964	96,177.4
19.00	2,025.0	4.00	77.00	0.5130	96,869.6
19.00	2,025.0	5.00	45.00	0.4216	59,975.2
19.00	2,025.0	6.00	72.00	0.4595	93,337.6
19.00	2,025.0	7.00	87.00	0.5054	122,896.7
19.00	2,025.0	8.00	80.00	0.4756	112,278.4
19.00	2,025.0	9.00	65.00	0.4598	119,792.6
19.00	2,025.0	10.00	57.00	0.4707	85,782.8
19.00	2,025.0	11.00	60.00	0.4544	90,876.5
19.00	2,025.0	12.00	45.00	0.4702	60,773.6

10. SKU portfolio

Top 20 SKUs by produced volume

sku	marca	familia	tipo_envase	n_ofs	uds_total	hl_total	oeo_medio
ED13LTW	ESTRELLA DAMM	130624PCAFCT00	LATA 1/3 SR.	132	96,904,066.0	318,707.5	0.5631
VO13LTNN	VOLL DAMM	130624PCAFCT00	LATA 1/3 SR.	77	67,468,389.0	217,588.7	0.5781
3BNZFLB1	LA FRÍA	130024PCTR0000	LATA 1/3 SR.	36	61,383,609.0	200,278.6	0.6261
3BNEBL23	EMD BRAU	130012PCDF0000	LATA 1/3 SR.	45	53,167,883.0	160,189.9	0.4705
ED13LTCW	ESTRELLA DAMM	130624PCAFCT00	LATA 1/3 SR.	69	51,573,593.0	163,776.9	0.5650
ED13P12M	ESTRELLA DAMM	131212MP00PK00	LATA 1/3 SR.	54	32,508,067.0	106,380.6	0.5979
ED12LTW	ESTRELLA DAMM	120624PCAFCT00	LATA 1/2 SR.	33	25,736,462.0	128,682.3	0.5586
XI13LTN	XIBECA	130624PCAFCT00	LATA 1/3 SR.	44	25,214,168.0	81,154.7	0.6107
ED13LP12	ESTRELLA DAMM	131212PC00PK00	LATA 1/3 SR.	55	23,198,048.0	76,572.5	0.5809
SK13LN	SKOL	130624PCAFCT00	LATA 1/3 SR.	35	19,513,436.0	61,985.8	0.5732
3BNHJLB2	SHEKELS	130024PCTR0000	LATA 1/3 SR.	8	17,504,462.0	57,764.8	0.7417
TU13LTN	TURIA MARZEN	130624PCAFCT00	LATA 1/3 SR.	43	14,384,472.0	45,709.9	0.5979
DL13LT	DAMM LEMON	130624PCAFCT00	LATA 1/3 SR.	23	12,542,979.0	41,391.9	0.5399
FDL13LN	FREE DAMM LIMON	130624PCAFCT00	LATA 1/3 SR.	40	11,684,245.0	38,558.0	0.5087
FDT13LT	FREE DAMM TOSTADA	130624PCAFCT00	LATA 1/3 SR.	44	11,596,183.0	37,103.0	0.5403
3BNULLB0	UNLAGUER	130024PCTR0000	LATA 1/3 SR.	6	11,122,089.0	36,702.9	0.6761
LC13LTNN	ESTRELLA LEVANTE	130624PCAFCT00	LATA 1/3 SR.	19	10,969,783.0	34,259.5	0.5556
VO13PL12	VOLL DAMM	131212PC00PK00	LATA 1/3 SR.	40	10,964,584.0	34,644.7	0.5693
FD13LTNN	FREE DAMM	130624PCAFCT00	LATA 1/3 SR.	32	10,865,862.0	34,917.9	0.5508
3BVMLLB0	Molen	130024PCTR0000	LATA 1/3 SR.	13	10,216,153.0	33,713.3	0.5130

Worst-performing SKUs (≥ 10 runs in 2025)

sku	marca	tipo_envase	n_ofs	oeo_medio
DL13LP4A	DAMM LEMON	LATA 1/3 SR.	16	0.2334
FD13LP4A	FREE DAMM	LATA 1/3 SR.	10	0.2856
CM13LT	COMPLOT	LATA 1/3 SR.	15	0.3018
SK13L12	SKOL	LATA 1/3 SR.	13	0.3103
SK1312MN	SKOL	LATA 1/3 SR.	19	0.3461
EN13LB24	ESTRELLA NON-ALCOHOLIC	LATA 1/3 SR.	10	0.3556
LC12LTW	ESTRELLA LEVANTE	LATA 1/2 SR.	23	0.3638
RB13LTN	ROSA BLANCA	LATA 1/3 SR.	14	0.3681
RB13L12N	ROSA BLANCA	LATA 1/3 SR.	15	0.3701
TNT13LT	TURIA MARZEN 0°0 TOSTADA	LATA 1/3 SR.	13	0.3735
EX13LTIB	ESTRELLA DAMM	LATA 1/3 SR.	10	0.3857
EX1224LB	ESTRELLA DAMM	LATA 1/2 SR.	13	0.3889
EN25LBUK	ESTRELLA NON-ALCOHOLIC	LATA 2/5	10	0.3897
VO13LTP6	VOLL DAMM	LATA 1/3 SR.	22	0.3978
ED13LCMM	ESTRELLA DAMM	LATA 1/3 SR.	27	0.3979
7BNS9LB1	SAN MARCOS	LATA 2/5	17	0.4001
VO13LTMN	VOLL DAMM	LATA 1/3 SR.	29	0.4075
EX1324BN	ESTRELLA DAMM	LATA 1/3 SR.	17	0.4084
7BNG3LB2	GRANDE	LATA 2/5	13	0.4136
VI12LTW	VICTORIA	LATA 1/2 SR.	16	0.4206

Most-frequent SKU pair transitions

linea	prev_sku	next_sku	prev_marca	next_marca	n	oeo_medio
19	LC12LTW	VI12LTW	ESTRELLA LEVANTE	VICTORIA	14	0.4414
19	ED13LP12	ED13P12M	ESTRELLA DAMM	ESTRELLA DAMM	12	0.6377
17	FD13LTNN	FDT13LT	FREE DAMM	FREE DAMM TOSTADA	11	0.6051
17	VO13LP24	VO13LTNN	VOLL DAMM	VOLL DAMM	10	0.6191
17	ED13LTCW	ED13LTW	ESTRELLA DAMM	ESTRELLA DAMM	10	0.6134
17	ED13LTW	ED13P24M	ESTRELLA DAMM	ESTRELLA DAMM	9	0.5212
14	ED13LTW	ED13LTCW	ESTRELLA DAMM	ESTRELLA DAMM	8	0.5166
19	FD13LP12	KE13PL12	FREE DAMM	KELER	8	0.5108
14	ED13LCMM	ED13LCMW	ESTRELLA DAMM	ESTRELLA DAMM	8	0.4384
19	VO13P12M	VO13PL12	VOLL DAMM	VOLL DAMM	8	0.5997
19	KE13PL12	TU13LP12	KELER	TURIA MARZEN	7	0.5460
17	ED13LTFW	ED13LTW	ESTRELLA DAMM	ESTRELLA DAMM	7	0.5817
14	VO13L6M1	VO13LTP6	VOLL DAMM	VOLL DAMM	7	0.4449
17	FDT13LT	VNT13LT	FREE DAMM TOSTADA	VICTORIA 0,0 TOSTADA	7	0.4518
14	ED13LTMW	ED13LTW	ESTRELLA DAMM	ESTRELLA DAMM	7	0.4484
14	SK1312MN	SK13L12	SKOL	SKOL	7	0.3410
17	TU13LTN	VO13LTNN	TURIA MARZEN	VOLL DAMM	6	0.6252
17	ED13LTCW	FD13LTNN	ESTRELLA DAMM	FREE DAMM	6	0.5014
17	ED13LTCW	FDL13LN	ESTRELLA DAMM	FREE DAMM LIMON	6	0.5562
19	TU13LP12	VO13PL12	TURIA MARZEN	VOLL DAMM	6	0.6651

11. Maintenance & LIMPIEZA events

OEE by number of maintenance calls during the OF

linea	llamadas_bucket	n	oee_medio
14	0	188	0.4256
14	1-2	173	0.4098
14	3-5	51	0.4364
14	6+	16	0.4543
17	0	465	0.5390
17	1-2	355	0.5206
17	3-5	89	0.5409
17	6+	25	0.5207
19	0	407	0.4815
19	1-2	295	0.4732
19	3-5	69	0.4946
19	6+	9	0.5524

Standalone LIMPIEZA WOs per line

linea	n_eventos	horas_total	horas_medio	horas_mediano	horas_intervencion_total	horas_espera_total
14.00	42.00	49.90	3.1200	1.4600	20.60	29.40
17.00	44.00	53.60	2.9800	1.2300	15.60	37.90
19.00	47.00	13.30	0.8900	0.6400	6.5000	6.8000

12. Plan vs Actual — May 2026 sample

The provided plan covers May 18–24 2026. Actual production data covers May 18–22 2026. Below: best-effort alignment on (linea, sku, date).

linea	sku	denominacion	fecha_ini_plan	fecha_fin_plan	start_ts_plan	turno_plan	cntd_jda	cntd_plan	secuencia_plan	of_actual	fecha_fin_actual
14.00	FDT13LT	FREE DAMM TOSTADA 1/3 LATA LC	2026-05-18	2026-05-18	2026-05-18 10:00:00	T	7,700.0	7,700.0	nan	PRT9900017389- M	2026-05-18
14.00	FDT13LT	FREE DAMM TOSTADA 1/3 LATA LC	2026-05-18	2026-05-18	2026-05-18 16:00:00	N	10,400.0	9,013.0	99.00	PRT9900017389- M	2026-05-18
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-19	2026-05-19	2026-05-19 08:00:00	T	10,500.0	10,500.0	nan	0000793838-1	2026-05-20
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-19	2026-05-19	2026-05-19 16:00:00	N	10,500.0	10,500.0	nan	0000793838-1	2026-05-20
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-20	2026-05-20	2026-05-20 00:00:00	M	10,500.0	10,500.0	nan	0000793838-1	2026-05-20
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-20	2026-05-20	2026-05-20 08:00:00	T	10,500.0	10,500.0	nan	0000793838-1	2026-05-20
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-20	2026-05-20	2026-05-20 16:00:00	N	8,400.0	8,400.0	nan	0000793838-1	2026-05-20
14.00	VO13LTMP	VOLL DAMM 33CL LT LC6 MP	2026-05-21	2026-05-21	2026-05-21 08:00:00	T	4,250.0	4,250.0	nan	0000798907-1	2026-05-20
14.00	VO13LTMP	VOLL DAMM 33CL LT LC6 MP	2026-05-21	2026-05-21	2026-05-21 10:00:00	N	3,000.0	3,000.0	nan	0000798907-1	2026-05-20
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-21	2026-05-21	2026-05-21 16:00:00	N	6,000.0	6,000.0	nan	0000793838-1	2026-05-20
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-22	2026-05-22	2026-05-22 16:00:00	N	10,500.0	10,500.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-22	2026-05-22	2026-05-22 08:00:00	T	10,500.0	10,500.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-22	2026-05-22	2026-05-22 00:00:00	M	10,500.0	10,500.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-23	2026-05-23	2026-05-23 16:00:00	N	9,975.0	9,975.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-23	2026-05-23	2026-05-23 08:00:00	T	9,975.0	9,975.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-23	2026-05-23	2026-05-23 00:00:00	M	9,975.0	9,975.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-24	2026-05-24	2026-05-24 16:00:00	N	9,975.0	9,975.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-24	2026-05-24	2026-05-24 08:00:00	T	9,975.0	9,975.0	nan	nan	nan
14.00	ED13LTNN	ESTRELLA DAMM 1/3 LATA LC	2026-05-24	2026-05-24	2026-05-24 00:00:00	M	9,975.0	9,975.0	nan	nan	nan
14.00	FDT13LTM	nan	nan	nan	nan	nan	nan	nan	nan	PRT9900017388- M	2026-05-18
14.00	ED13LMCM	nan	nan	nan	nan	nan	nan	nan	nan	PRT9900017390- M	2026-05-18
14.00	ED13LMCM	nan	nan	nan	nan	nan	nan	nan	nan	PRT9900017387- M	2026-05-18
17.00	XI13LTN	XIBECA 1/3 LATA LC	2026-05-18	2026-05-18	2026-05-18 08:00:00	T	12,327.0	12,327.0	nan	PRT9900017407- M	2026-05-18
17.00	XI13P24M	XIBECA 1/3 LATA PACK	2026-05-18	2026-05-18	2026-05-18 06:00:00	T	5,000.0	4,420.0	99.00	PRT9900017406- M	2026-05-18

linea	sku	denominacion	fecha_ini_plan	fecha_fin_plan	start_ts_plan	turno_plan	cntd_jda	cntd_plan	secuencia_plan	of_actual	fecha_fin_actual
		CARTON 24U. MP									
17.00	XI13LTN	XIBECA 1/3 LATA LC	2026-05-18	2026-05-18	2026-05-18 14:00:00	N	8,273.0	8,273.0	nan	PRT9900017407- M	2026-05-18

13. Daily HL planning — Programa Prod. vs Acordado

From the [Diario HL Planif.xlsx](#) daily replanning report (May 18–24 2026). *delta_hl* = Acordado – Programa Prod. Positive = added volume, negative = removed. Together with the format/quantity-modification counters, this is the **real-world ground truth** of how much a plan has to flex during the week.

linea	fecha	n_skus	prog_prod_hl	prog_acord_hl	delta_hl	total_cambios	total_modif_formato	total_modif_cantidad	total_inclusiones	total_exclusiones
14	2026-05-18	4	2,399.8	2,399.8	0.00	nan	nan	nan	nan	nan
14	2026-05-19	1	1,663.2	1,663.2	0.00	nan	nan	nan	nan	nan
14	2026-05-20	1	2,328.5	2,328.5	0.00	nan	nan	nan	nan	nan
14	2026-05-21	2	1,049.4	1,247.4	198.00	2.00	1.00	1.00	1.00	nan
17	2026-05-18	5	3,623.2	3,623.2	0.00	nan	nan	nan	nan	nan
17	2026-05-19	3	4,389.3	4,428.9	39.60	nan	nan	nan	nan	nan
17	2026-05-20	3	1,960.4	1,960.4	0.00	nan	nan	nan	nan	nan
17	2026-05-21	4	4,417.8	4,417.8	0.00	nan	nan	nan	nan	nan
19	2026-05-18	5	10,486.9	10,486.9	0.00	nan	nan	nan	nan	nan
19	2026-05-19	5	14,932.5	14,972.1	39.60	nan	nan	nan	nan	nan
19	2026-05-20	2	14,704.8	15,088.8	384.00	2.00	nan	2.00	nan	nan
19	2026-05-21	9	11,911.3	11,725.2	-186.00	6.00	3.00	3.00	1.00	2.00

DEMO GOLD

L19 had +384 HL added on May 20 and –186 HL removed on May 21 — with **6 changes** on that single day (3 format mods, 3 quantity mods, 1 inclusion, 2 exclusions). Use this week as the live scenario in the demo: replay the plan, inject those real interventions, watch LineWise rebalance.

14. Run-length distribution & CIP impact

Hours per OF — distribution by line

linea	n_ofs	h_min	h_p25	h_mediana	h_p75	h_max
14.00	436.00	0.00	5.1800	8.1700	16.28	86.96
17.00	950.00	0.00	2.8100	5.0200	9.5600	50.86
19.00	792.00	0.00	3.0100	4.7600	8.9700	96.29

OEE bucketed by CIP duration

linea	cip_bucket	n	oeo_medio
14	cip_0	410	0.4257
14	cip_30_90min	3	0.4223
14	cip_gt_90min	9	0.2652
14	cip_lt_30min	6	0.3740
17	cip_0	900	0.5370
17	cip_30_90min	12	0.3626
17	cip_gt_90min	8	0.2499
17	cip_lt_30min	14	0.4992
19	cip_0	754	0.4835
19	cip_30_90min	4	0.3258
19	cip_gt_90min	9	0.2434
19	cip_lt_30min	13	0.5091

15. Best & worst runs by OEE

kind	of	linea	fecha_fin	sku	marca	oeo	horas_totales	horas_cip
best	0000570206-1	17	2025-04-22	FD13LN	FREE DAMM	1.5730	1.4606	0.00
best	PRT9900015467-M	19	2025-02-13	ED13P12M	ESTRELLA DAMM	0.9234	0.7039	0.00
best	0000589724-1	17	2025-05-28	XI13LP12	XIBECA	0.9027	0.9189	0.00
best	0000543295-1	19	2025-02-21	3BNULB0	UNLAGUER	0.8691	51.55	0.00
best	0000676483-1	19	2025-10-17	3BNZFLB1	LA FRÍA	0.8631	31.11	0.00
best	0000530634-1	19	2025-01-24	3BNZFLB1	LA FRÍA	0.8502	14.35	0.00
best	0000667570-1	19	2025-09-28	3BNZFLB1	LA FRÍA	0.8465	96.29	0.00
best	0000620488-1	19	2025-07-19	3BNZFLB1	LA FRÍA	0.8433	29.35	0.00
best	0000535069-1	19	2025-02-07	VO13P12M	VOLL DAMM	0.8405	3.2000	0.00
best	PRT9900016778-M	17	2025-12-03	VO13LP24	VOLL DAMM	0.8376	3.7094	0.00
worst	0000574254-1	17	2025-04-28	VO13P12M	VOLL DAMM	0.0998	0.7461	0.00
worst	PRT9900015618-M	14	2025-03-24	ED13LTW	ESTRELLA DAMM	0.0845	1.6667	0.00
worst	0000694994-1	14	2025-11-11	3BNZFLB1	LA FRÍA	0.0780	1.0725	0.00
worst	0000660750-1	19	2025-09-17	ID12LBN	INEDIT	0.0631	9.4658	2.9697
worst	PRT9900015627-M	17	2025-03-28	OL13LTN	ORO	0.0586	3.3425	1.4511
worst	0000689263-1	17	2025-11-03	ED13LTCW	ESTRELLA DAMM	0.0491	5.2844	0.00
worst	0000689263-1	17	2025-11-03	ED13LTCW	ESTRELLA DAMM	0.0491	5.2844	0.00
worst	0000530603-1	14	2025-01-27	ED13LTW	ESTRELLA DAMM	0.0120	0.5767	0.00
worst	0000589707-1	17	2025-06-04	FDL13LN	FREE DAMM LIMON	0.0016	0.5119	0.00
worst	0000684006-1	19	2025-10-23	CM13LT	COMPLIT	0.0015	3.9361	0.00

16. Data quality notes

#	Issue	Mitigation
1	<code>OEE > 1</code> on ~12 rows (e.g. 1.57 on a FREE DAMM run on L17). Likely a unit / aggregation glitch.	Clip to <code>[0, 1]</code> in modelling. Flag when <code>> 1</code> .
2	<code>H. Tot.</code> outliers (max 21,065 h on one row).	<code>fact_runs.outlier</code> filters <code>H. Tot. > 100h</code> .
3	LIMPIEZA OFs have NaN OEE.	Split into <code>fact_limpieza</code> ; excluded from <code>fact_runs</code> .
4	41 OFs in OEE file missing from Cambios file.	LEFT JOIN; dimension flags become NULL.
5	Date granularity = <code>Fecha Fin</code> only (no time-of-day for OFs).	Sort by <code>(fecha_fin, of)</code> to get a stable order.
6	May 2026 plan duplicates rows per shift (T/N/M).	<code>fact_plan_vs_actual_2026.estado_join</code> ∈ {matched, only_plan, only_actual}.
7	Some <code>C.*</code> dimension columns contain large ints (163, 803, etc.) instead of 0/1.	Use the <code>c_*_flag</code> boolean columns (engineered: any value <code>> 0</code> → TRUE).
8	Cambios file uses <code>Nº</code> → ingested as <code>no_</code> (e.g. <code>no_llamadas</code> , <code>no_de_cambios</code>).	Column-name normalisation: lowercase ASCII snake_case.

17. Implications for the LineWise build

- **Sequencer objective:** minimise the number of *size* changes first, then *brand/producto/cap* changes. Use the empirical OEE deltas as weights (volum: -10.7, producto: -5.8, brand: -5.7, cap: -5.5, primario: -3.9, secundario: -3.4, palet: -2.6).
- **OEE estimator features** (already in `fact_runs` ready to use): `linea`, `sku`, `prev_sku`, `cambio_tipo_principal`, `estado_volumen`, `prev_estado_volumen`, `dia_semana`, `mes`, `n_llamadas_mant`, `horas_cip`, plus the 8 `c_*_flag` booleans.
- **Target variable:** `fact_runs.oee`. Clip to `[0, 1]` before training.
- **Diagnostic panel:** rank by `SUM(minutos)` from `fact_lost_time`, grouped by `(linea, categoria)`. The eight categories are ready-made.
- **Time Machine backtest:** split `fact_runs` at week 40 of 2025; the last ~13 weeks become the holdout. Aggregate "predicted vs actual OEE" gives the killer slide number.
- **Claude tool schema** should match the canonical tables: `query_history(linea, sku, prev_sku, window) → fact_runs` aggregations; `find_analogs(of_id, k) → kNN` on encoded feature vector; `score_sequence(plan) →` empirical lookup + LightGBM blend; `list_diagnostics(linea, week) → fact_lost_time` aggregation.

BOTTOM LINE

The data is richer than the brief suggested. We have lossless OEE history, an already-decomposed lost-time breakdown, a theoretical changeover matrix, and a plan-vs-actual May 2026 sample for the validation story. There is no need to invent features: the dataset hands us the answer to "why did OEE drop?" — we just need to surface it intelligently and let the optimizer choose better sequences.